Tran Thanh Gia Khang

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Summary

Information Technology student specializing in Artificial Intelligence with hands-on experience in machine learning pipelines, feature engineering, model evaluation, and application deployment. Interested in applying ML to healthcare and agriculture. Seeking an AI/ML Engineer Intern position to develop and evaluate practical ML solutions.

Education

FPT University **Sep. 2024 – Dec. 2028 (Expected)**
Bachelor of Science in Information Technology (Artificial Intelligence) *Can Tho, Vietnam*

- **Relevant Coursework:** AI, Data Science with Python & SQL (8.7/10); AI, Data Science Project (8.4/10); Machine Learning (8.3/10).
- **Activities & Leadership:** Team Captain (Hovilo Games Esports Team): Led a team of 5 players, coordinated strategies, and secured 2nd place in the 2026 Hovilo Games Esports Tournament.

Technical Stacks

Programming Languages: Python, SQL.
Machine Learning: Scikit-learn, XGBoost, Feature Engineering, Model Evaluation, Stacking Ensemble.
Computer Vision: OpenCV, scikit-image, HOG, CLAHE, HSV Color Histograms.
Data Processing & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Plotly.
Tools & Deployment: Git, GitHub, Streamlit, Jupyter Notebook, Kaggle, VS Code.
Soft Skills: Analytical Thinking, Problem Solving, Team Collaboration, Time Management.

Projects

Cardiovascular Disease Risk Prediction | Group Project GitHub | Live demo

- **Role:** Led end-to-end ML system design and implementation in a 4-member team, including model development, feature engineering, XGBoost tuning, Streamlit deployment, and a TF-IDF/cosine similarity-based chatbot.
- Built an ML pipeline on a **308K-row** healthcare dataset, applying ordinal encoding and engineering three features: BMI category, comorbidity count, and unhealthy diet score.
- Handled class imbalance using XGBoost's scale_pos_weight parameter and benchmarked multiple classifiers, achieving an **ROC-AUC of 0.816** and 69% recall on the positive class.

Rice Leaf Disease Classification | Individual Course Project GitHub

- **Built** an end-to-end classical ML pipeline (no deep learning): **CLAHE** preprocessing, then 198 hand-crafted features from **HSV color histograms and HOG**.
- **Applied** OpenCV augmentation (flip, brightness, rotation $\pm 10^{\circ}$, blur) to the training set only, after the train/validation split to avoid leakage, expanding 1,722 images to 12,054 samples.
- **Compared** five classical models and a **Stacking Ensemble** (RF, SVM, KNN), achieving a **Macro F1 of 0.957** on the local validation set (304 images, 8 classes) and a **Kaggle private score of 0.782** on the hidden test set.

Vietnamese Dense Retrieval: Web Corpus & Evaluation | Group Project GitHub

- **Role:** Retrieval and evaluation in a team of 5, supporting hard negative mining for Vietnamese dense retrieval; reviewed **400 relevance labels** for the UEL Admissions FAQ corpus.
- **Compared TF-IDF and BM25** across 4 preprocessing variants on 40 queries; baseline **BM25** performed best (**MRR@5 0.988, nDCG@5 0.985, Hit@1 0.975**) versus 0.950, 0.954 and 0.925 for TF-IDF.
- **Found** that stopword removal, stemming and lemmatization did not improve over the baseline, using Precision@5, Recall@5, Hit@1, MRR@5 and nDCG@5.

Certifications

IBM Data Science Professional Certificate July 2026
Coursera — Credential
Ethics in the Age of AI Specialization April 2026
Coursera — Credential
AI Foundations for Everyone Specialization December 2025
Coursera — Credential

Languages

English: Working proficiency (technical reading and writing) — **Vietnamese:** Native.